Sampling more does not help, because the bottleneck is choosing:
a measured account of the identifiability gap at 3B

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Abstract

A self-consistency system fails in two ways: it never generates the right answer, or it generates it and does not pick it. On 1,000 held-out math word problems with a fixed 3B base model at $k = 64$, plurality voting scores 0.8250, at least one sample is correct on 0.9250, so the selection failure is worth 0.100 and the generation failure 0.075. We price the selection half by injecting a synthetic verifier of controlled quality into the real generation and applying the real fusion rule: recovering half the selection loss takes a within-problem pairwise AUC of 0.773, three quarters takes 0.883, and full recovery is out of reach at any weight. We built nine selectors — a trained outcome reward model, four prompted judges, length-normalised log-probability, and two fine-tuned generators used as scorers. Our best reaches an AUC of 0.7545 overall and 0.6599 where the answer is decided, yet delivers what a homogeneous verifier of AUC **0.578** would: 7% of the selection loss against the 46% its measured quality implies. The gap is not weakness but shared error — a reward model adapted from the generator's own weights fails on the problems the generator fails on, which a global correlation of 0.18 does not reveal. Fusing that verifier with the vote is worth +0.0108 on a held-out leaderboard set against plurality voting alone, averaged over three independent generations spanning +0.0084 to +0.0120: 10.8% of the selection loss, and the only gain we collected. We then ran seven preregistered interventions to collect more — four that widen the sample pool, two that improve the selector, and one that gives the verifier the enlarged pool to choose from. Each moved the quantity it targeted; none reached two standard errors on submitted accuracy, and the condition with the highest coverage produced the lowest accuracy. Seven LoRA training runs, spanning supervised distillation from a stronger teacher, preference optimisation, and reward modelling, left per-sample accuracy at or below the untrained base. Two annotators independently classifying 30 of the 75 problems no sample solves find 20.0% not solvable as posed; four gold answers are wrong by computation and on two of them the model's modal answer is the correct one. The reachable ceiling is 0.940, not 1.000. The standard portfolio result for combining weakly correlated signals overpredicted our measured combination gains by factors of 8.2 and 6.6. We report the preregistration record in full, including three null results of our own that a larger sample later reversed, and the public leaderboard rank change produced by re-running one configuration without changes.

1 Sampling more does not help, because the bottleneck is choosing

A self-consistency system answers a math problem twice over. It samples k solution chains, and it picks one answer from them. Both steps fail, and they fail for different reasons, but the failures arrive in the same place: a wrong integer in a submission file.

We ran a competition system under a fixed base model — Qwen2.5-3B-Instruct, whose weights we were not permitted to replace — and measured the two failures separately on 1,000 held-out problems at $k = 64$:

$$\underbrace{0.8250}_{\text{maj@64}} \xrightarrow{0.100} \underbrace{0.9250}_{\text{pass@64}} \xrightarrow{0.075} 1.000. \tag{1}$$

Ten points of the gap to perfect accuracy sit on problems the model already solves at least once and then discards. Seven and a half points sit on problems it never solves. The larger share is a selection failure, and selection is the cheaper of the two to attack: it needs no additional inference.

We attacked it, and we failed. This paper reports the failure in a form that is usable, because we can say how far short we fell and why the shortfall is structural rather than a matter of tuning.

The requirement. Our fusion rule scores a candidate answer a by $s(a) = \log n_a + \gamma\, z(a)$, where n_a counts the samples producing a and z standardises a verifier score within the problem. Overturning a plurality-wrong problem requires the correct candidate to overcome its vote deficit. Across plurality-wrong problems the median deficit corresponds to a separation of $\Delta z > 2.2\sigma$, equivalently a within-problem pairwise AUC of 0.940. This number is the paper's measuring stick. We built nine scorers. The best reaches 0.583σ, an AUC of 0.660: a factor of 3.8 short.

Five interventions, one result. Each intervention moved the quantity it was designed to move, and none of them moved the score.

Axis	Intervention	Basis	target metric	submitted accuracy
generation	checkpoint pooling	earlier	pass +0.007 to +0.013	0.000
generation	temperature pooling	earlier	pass +0.010	negative
generation	prompt pooling	primary	pass +0.0060 (matched)	−0.0040 (1.1 SE)
generation	repeated-run pooling	primary	pass +0.0070 (2.7 SE)	+0.0020 (1.0 SE)
selection	verifier stance	primary	AUC +0.0359 (3.2 SE)	+0.0060 (1.9 SE)

Four of these widen the pool of hypotheses; the fifth improves the choice among them. The requirement above explains both columns. A wider pool adds correct answers the selector cannot find, and a selector at a quarter of the needed separation cannot find them. Seven LoRA training runs — supervised distillation from a stronger teacher, preference optimisation, and reward modelling — produced the same outcome, at per-sample accuracies at or below the untrained base.

Measurement. Effects of this size are only meaningful next to the noise floor, so we measured that too. Rerunning one configuration unchanged, with the same seed, changes 42 of 1,000 validation answers and 14 of 831 leaderboard answers; aggregate accuracy moves by 0.002, and the repeat submission landed one rank lower on the public leaderboard than the original. We state reproducibility as ± 0.004 from that direct measurement, and we report an earlier estimate of ± 0.016 as an error, because the two numbers license different conclusions about everything else in the paper.

Contributions.

1. A decomposition of self-consistency error into selection loss (0.100) and generation loss (0.075), computable from a single generation file, that partitions the design space and prices each half.

2. A measured price for the selection half, obtained by injecting a synthetic verifier of controlled quality into the real generation: half the selection loss costs an AUC of 0.773, three quarters costs 0.883.

3. The decomposition of one verifier's shortfall into three stacked deficiencies — insufficient AUC, weakness concentrated where the answer is decided, and errors that coincide with the generator's. The third is the largest: our model measures 0.7545 and behaves like 0.578.

4. Five interventions, preregistered and measured with paired bootstrap intervals, each significant on its target metric and none on accuracy — the same failure on both axes, and evidence that the identifiability gap (Bay and Yearick, 2026) closes on neither.

5. A measured null for the standard combination result of Grinold (1989) in this setting: predicted gains exceed measured gains by factors of 8.2 and 6.6.

6. A twice-annotated classification of the problems no sample solves. Of 30 examined, 20.0% are defective rather than hard, with four gold answers proved wrong by computation. The reachable ceiling is 0.940. The two annotators agree on 24 of 30 ($\kappa = 0.68$) and every disagreement runs the same way, so the rate is a floor.

7. A preregistration and self-correction record, including three of our own null results that a larger sample later reversed, and the leaderboard rank change produced by re-running an identical configuration.

One thing in this paper raised the submitted score, and we can say exactly how much. On one generation, plurality voting alone scores 0.7882 on the public leaderboard and adding the verifier at $\gamma = 0.5$ scores 0.7966. Two further generations, each scored against its own plurality control, give +0.0120 and +0.0120; the three measurements average +0.0108, which is 10.8% of the selection loss the decomposition identifies, and it is the whole of what we collected. Every subsequent attempt — four pooling methods, a retrained verifier, a second independent scoring signal, and seven training runs — moved the submitted score by at most one problem in 831. The paper's contribution is not the 10.8%; it is the measured account of why the remaining 89.2% stayed out of reach, in a form that transfers: before spending compute on either axis, measure the split, and price the selector the split demands.

2 Prior work names the gap; this paper prices it

Two quantities, one gap. Self-consistency answers a problem by plurality vote over sampled chains of thought (Wang et al., 2023). Repeated sampling raises coverage — the fraction of problems with at least one correct sample — far faster than it raises the accuracy of the chosen answer (Brown et al., 2024). The space between the two curves is the object of this paper. Bay and Yearick (2026) name it the identifiability gap and show that selection saturates within dozens of samples while coverage keeps rising, so that further sampling can lower accuracy. Their result says the gap exists and does not close by sampling. It does not say what closing it would cost. We convert the gap into a requirement on a selector — a separation of 2.2σ between correct and incorrect candidates on plurality-wrong problems, equivalently a within-problem pairwise AUC of 0.940 — and measure five selectors against it on one basis.

Verifiers, and their limits. The standard instrument for the selection step is a trained verifier: an outcome-supervised reward model (Cobbe et al., 2021), a process-supervised one (Lightman et al., 2024), or a compute-allocation policy built on either (Snell et al., 2024). Stroebl et al. (2024) establish that an imperfect verifier bounds what resampling can deliver, and that under realistic false-positive rates the optimal sample count falls below ten. Their bound is asymptotic in k and expressed through false positives. Ours is a fixed-budget statement: at $k = 64$, under an explicit fusion rule, a verifier must separate correct from incorrect candidates by 2.2σ to be worth more than the vote it overrides. The best verifier we trained reaches 0.583σ. The distance is a factor of 3.8, not a matter of tuning.

Combining weak signals. When several imperfect signals are available, portfolio theory prescribes combining them, and the gain grows as their correlation falls (Grinold, 1989). We measured the inputs the formula requires and the output it predicts. Predicted gains exceeded measured gains by factors of 8.2 and 6.6. We report the discrepancy as a measured property of this setting rather than a defect of the theory, and draw the operational consequence in §5.

Reporting practice. Effects in this paper are between 0.003 and 0.013, and our own repeated runs of a single configuration move accuracy by 0.002. At that ratio, reporting conventions are part of the result. We follow the reproducibility program of Pineau et al. (2021) for environment and artifact disclosure, and add a preregistration record: every experiment's gate, trial count, reading rule, and the author's prediction, written before the measurement existed (§10). The record is what allows us to report three of our own null results as underpowered rather than as findings.

3 Setup: a fixed 3B base, 64 samples per problem, 1,000 validation problems

3.1 Task and metrics

The task is a competition benchmark of mathematics word problems in English whose answers are integers. Scoring is exact match; a blank answer counts as wrong. The training split holds roughly 25k problems with gold answers, a public leaderboard split holds 831 problems, and a held-out test split of about 2000 problems is released on the final day.

Three quantities carry the paper, all computed from a single generation of k samples per problem.

Metric	Definition	Role
sample accuracy	correct samples / all samples, pooled	the solver
maj@k	accuracy of the plurality answer	what the system submits
pass@k	problems with ≥ 1 correct sample	ceiling of any selector

Table 1: Ties in maj@k break deterministically to the smallest answer, so the metric does not depend on sample order.

Answers are extracted with a fixed parser: the last `\boxed{}` expression when present, otherwise the first number after a closing phrase, otherwise the last number on the final line. Only exact integers are accepted; no rounding.

One caveat belongs here rather than in a footnote. An earlier, boxed-only variant of this parser was used for the acceptance gates while the project ran, and it scores pass@64 at 0.9240 against the full parser's 0.9250. Every headline number in this paper uses the full parser. Where a gate threshold or a gate outcome is quoted, the boxed-only figure is the one that was in force at the time, and we quote it as such.

3.2 Constraints

	Rule
Fixed	base model `Qwen2.5-3B-Instruct`
Forbidden	loading or merging weights from any other model
	calling external models at inference time
	pretraining from scratch
	tool use and code execution at inference time
	sending test problems to any external service
Permitted	majority voting, self-consistency, best-of-n
	LoRA adapters on the fixed base, ensembled at inference
	public datasets, and training data built with commercial APIs

Table 2: Competition constraints. Inference runs offline.

These constraints shape the paper. They rule out the two levers with the largest known effect on this task — a larger model and a tool-using solver — and leave sampling, selection, and low-rank adaptation. That is why the decomposition in §4 is the natural frame: it partitions exactly the space the rules leave open.

3.3 Validation protocol

We hold out 1000 problems from the training split as an internal validation set and use it for every gate and every ablation. We verified that this set has empty intersection with all five training files used anywhere in the project, and with the quarantine set described below.

The organisers published a list of 627 training problems with erroneous labels, which we exclude. We additionally quarantined 440 problems flagged by an internal consistency check; this set is disjoint from the official list and from validation. A further set of mislabelled and ill-posed problems was acknowledged by the organisers during the competition but never distributed, so it remains in the data.

3.4 Base system

The configuration used for every measurement, and submitted unchanged, is the base model with no adapter, one fixed prompt with the chat template applied, $k = 64$ samples at temperature 0.8 with a 3072-token budget, and a fusion rule $s(a) = \log n_a + 0.5\, z(a)$ over a trained reward model. The reward model is the only trained component in the final system. It is a LoRA adapter on the same base, trained on solutions the base model generated for training problems — no external corpus and no teacher output enters the submitted system.

4 Selection loss is 0.100; generation loss is 0.075, and a fifth of it is unsolvable

A self-consistency system answers a problem in two steps. It draws k samples, and it picks one answer from them. Two different failures end the same way, and separating them tells us where effort can still pay.

Let maj@k be the accuracy of the plurality answer over k samples and pass@k the fraction of problems for which at least one of the k samples is correct. On our validation set,

$$\underbrace{0.8250}_{\text{maj@64}} \xrightarrow{+0.100} \underbrace{0.9250}_{\text{pass@64}} \xrightarrow{+0.075} 1.000. \tag{2}$$

We call the first interval the *selection loss* and the second the *generation loss*. The selection loss covers problems the model already solves at least once but does not choose; a perfect verifier would collect all of it. The generation loss covers problems no sample answers correctly; no verifier can reach them. Figure 1 shows the split.

The decomposition has two properties. It is measurable without any additional inference, since both quantities come from one generation file. And it partitions the design space: a change either widens the pool of hypotheses or improves the choice among them, and the decomposition says how much each is worth.

How large must a verifier be to collect the selection loss? Our fusion rule scores a candidate answer a by $s(a) = \log n_a + \gamma\, z(a)$, where n_a is the number of samples that produced a and z is the verifier score standardised within the problem. For the verifier to overturn a plurality-wrong problem, the correct answer must overcome the vote deficit. Across plurality-wrong problems the median log-count deficit corresponds to a required separation of $\Delta z > 2.2\sigma$ at our operating γ, equivalently a within-problem pairwise AUC of 0.940. The best of our nine scorers delivers 0.583σ, an AUC of 0.660: a factor of 3.8 short. The requirement scales as $1/\gamma$, so it is a joint statement about the verifier and the weight it is given; §5 sweeps γ and shows that every setting which relaxes the requirement enough to satisfy this verifier also scores below plurality voting alone.

Scaling k widens the gap rather than closing it. Figure 2 plots both quantities against k on a pooled set of 128 samples per problem. From $k = 1$ to $k = 128$, coverage rises by 0.204 and selection accuracy by 0.099. The gap grows monotonically and shows no sign of saturating. Sampling more does not make the choice easier; it makes the pool of unchosen correct answers larger.

4.1 Sampling more does not close the gap, out to 256

Figure 2 runs to $k = 128$ on a pooled generation. A separate 256-sample generation of 300 of the same problems lets us look further.

k	8	16	32	64	128	256
maj@k	0.7934	0.7998	0.8061	0.8079	0.8086	0.8133
pass@k	0.8719	0.8978	0.9180	0.9315	0.9418	0.9533
gap	0.0785	0.0980	0.1119	0.1236	0.1332	0.1400

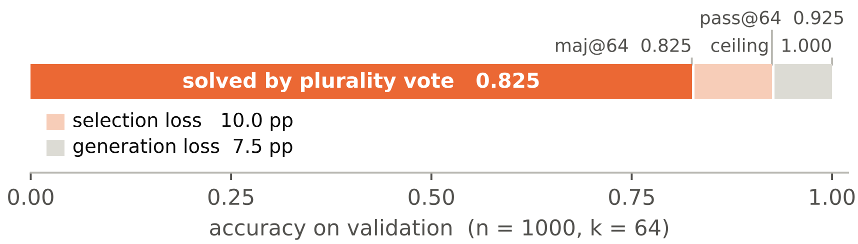

Figure 1: Selection loss (0.100) exceeds generation loss (0.075). Both quantities come from one generation file: maj@64 is the accuracy of the plurality answer, pass@64 the fraction of problems with at least one correct sample. The left interval is collectable by a selector; the right interval is not.

From $k = 8$ onward each doubling of the sample budget buys between 0.002 and 0.006 of plurality accuracy and between 0.010 and 0.026 of coverage. The gap widens at every step and is still widening at 256. The last doubling, from 128 to 256, adds 0.0115 of coverage and 0.0048 of accuracy: the run costs twice the compute and delivers a change indistinguishable from the run-to-run variation of §9. (These figures are on the 300-problem subset for which a 256-sample generation exists, so they are not directly comparable to the 1,000-problem numbers elsewhere; the shape is what matters.)

4.2 One in five uncovered problems is not solvable as posed

The generation loss is 0.075: 75 of 1000 validation problems receive no correct sample in 64 attempts. We characterise that set, because its composition decides whether the loss is worth attacking.

Three aggregate facts come first. Parsing is not the obstacle — 96.5% of samples on uncovered problems yield a well-formed integer. The model does not converge on a wrong answer; it scatters. Across uncovered problems the median number of distinct answers among 64 samples is 32, and the median share of the most frequent answer is 0.203. Only 9 of 75 problems concentrate at least half their samples on one wrong answer.

We then sampled 30 of the 75 (seed fixed in advance) and classified each by hand. To put a number on how much that classification depends on the annotator, a second annotator repeated it independently, without access to the first labels, under the same category definitions and with the same instruction to verify computationally wherever possible.

The disagreements are not noise around a shared judgement. Four of the six are problems the first annotator called hard reasoning and the second explained with a named misreading — dropping "four or more" from a consecutive-sum condition, dropping the leading-zero constraint, reading a connectivity requirement as a spanning tree. Two are gold answers the second annotator proved wrong by computation and the first did not check. **A single annotator under-counts even the class that is decidable by arithmetic**, which is the class this benchmark most needs counted.

Four gold answers are wrong, each proved by computation.

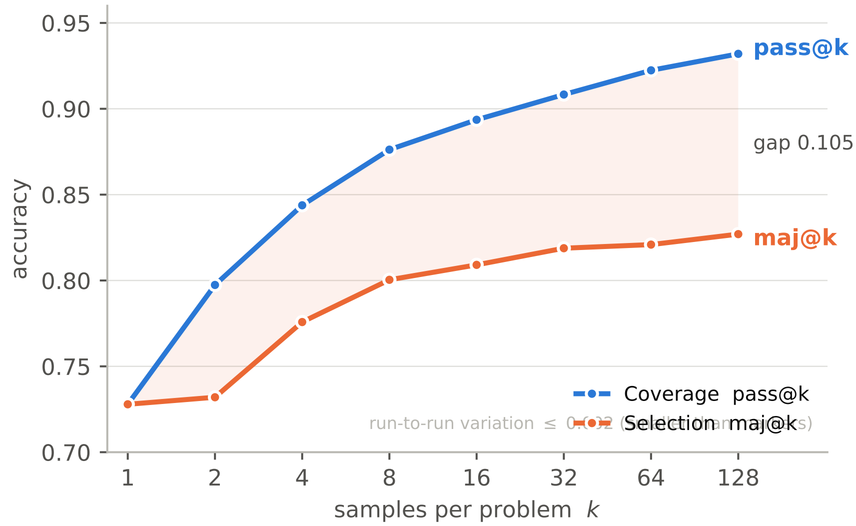

Figure 2: From $k = 1$ to $k = 128$, coverage rises by 0.204 and plurality accuracy by 0.099. The gap widens monotonically. Sampling more enlarges the set of correct answers the system does not choose.

Problem	Quantity	gold	computed
$\sqrt[4]{52200625}$	$85^4 = 52{,}200{,}625$; $51^4 = 6{,}765{,}201$	51	85
$1234 \cdots 1238 \bmod 17$	direct evaluation	9	13
order of $\prod_{n \le 10} a_n$, a_n the product of first-quadrant roots of $x^{2n} = 1$	argument $= 883/1260$ turn, $\gcd(883, 1260) = 1$	315	1260
cube of side 8, $d = (p - \sqrt{q})/u$, $p + q + u$	$d = (27 - \sqrt{186})/3$	196	216

On the first two the model's most frequent answer is the correct one — 85 on 15 of 64 samples, 13 on 35 of 64 — so the system is scored as failing completely on problems it in fact solves. The other two it also fails, but the score it is given is not the score the problem deserves.

A third problem fixes $\angle ABC = 90°$, $AB = 100$, $BC = y$, $AC = 2y - 10$, and a perimeter of 290. The perimeter gives $y = 200/3$, at which $AB^2 + BC^2 = 130000/9$ and $AC^2 = 136900/9$. The constraints are inconsistent; the triangle does not exist. A fourth carries corrupted mathematical markup that no reading resolves.

Consequence for the ceiling. Applying the second annotator's rate to the uncovered set, roughly 15 of the 75 problems are not solvable as posed (interval $[7, 28]$). The reachable ceiling is therefore near 0.9400 rather than 1.000, and the generation loss available to any method is 0.060 rather than 0.075 — a fifth of what the raw number suggests is not there to win.

An external check on the rate. The organisers inspected the 1,000-problem evaluation set in full and removed 169 of them — 16.9% — for wrong gold answers, missing figures, or answers that are not integers. We did not have that number when we ran our own inspection, and it is the closest thing to ground truth available for the defect rate in this benchmark family. Our hand-checked rate on uncovered problems is 20.0%, interval $[9.5\%, 37.3\%]$, which contains it.

Category	annotator 1	annotator 2
Defective item: gold answer wrong	2	4
Defective item: statement contradictory or corrupted	2	2
Composite answer encoding	3	3
Multi-part question, single integer field	1	1
Systematic misreading, convergent	2	5
Competition-level reasoning, dispersed	20	15
Total	30	30
Defective (wrong gold + broken statement)	13.3%	20.0%

Table 3: Thirty uncovered problems, classified independently twice. Agreement is 24/30, Cohen's $\kappa = 0.68$. Every one of the six disagreements runs the same way: the second annotator identified a defect or a specific misreading where the first defaulted to "hard". The defective rate of 20.0% carries a Wilson interval of $[9.5\%, 37.3\%]$.

The training split tells a different story. There the organisers published a list of 627 problems to exclude out of roughly 25,000: 2.5%, against 16.9% on the set they inspected exhaustively. Sampling 40 flagged candidates from what remains and checking each by hand, we estimate a residual label-error rate of 2.6%, interval $[1.3\%, 4.9\%]$, and we verified six errors by direct computation — among them a fourth root, a modular product, and a count of integers in an interval, each of which the model's most frequent answer got right and the gold label got wrong. Filtering that removes 2.5% from a split whose exhaustively inspected sibling yields 16.9% leaves roughly its own size behind. Our sample is consistent with that arithmetic.

Split	Inspection	Defect rate	Source
Evaluation, 1,000	exhaustive, by organisers	16.9%	169 removed
Training, $\approx 25{,}000$	partial, by organisers	2.5%	627 listed
Training, residual	40 flagged, hand-checked	2.6%	this work, $[1.3, 4.9]$
Uncovered validation, 75	30 sampled, two annotators	20.0%	this work, $[9.5, 37.3]$

Table 4: Defect rates by split and by how thoroughly each was inspected. The two exhaustive-inspection figures are an order of magnitude above the partial one.

Consequence for the two axes. The remaining 66.7% are competition problems — olympiad and invitational sources, not the word problems the task is described as. They are the dispersed majority: the model does not hold a wrong belief about them, it holds no belief. That distinction matters for the selection axis. A selector can only choose among generated candidates, and on these problems the candidates carry no correct answer to choose. It also refutes, for the third time in this project, the hypothesis that label noise dominates the unsolved set: defective items explain 20% of it, with a Wilson upper bound of 37%.

Scorer	AUC, all	AUC, plurality-wrong	Δz	shortfall
ORM, neutral stance	0.7545 ± 0.0113	0.6599 ± 0.0285	0.583σ	3.8×
ORM, adversarial stance	0.7456 ± 0.0112	0.6240 ± 0.0292	0.447σ	4.9×
DPO adapter	0.6110 ± 0.0083	0.5433 ± 0.0235	0.154σ	14.3×
teacher-SFT adapter	0.6042 ± 0.0077	0.5387 ± 0.0216	0.137σ	16.0×
ORM, mismatched prompt	0.4933 ± 0.0079	0.4992 ± 0.0153	-0.003σ	—
requirement	—	0.9401	2.200σ	1.0×

Table 5: Five scorers on the primary basis: 1,000 problems, $k = 64$, one generation file. The plurality-wrong column is computed on the 100 problems where plurality voting fails *and* at least one sample is correct; on the other 75 no selector can help, so no pairwise comparison exists. Intervals are paired bootstrap over problems, 2000 resamples. The requirement row is derived in §4.

5 The best verifier reaches one quarter of the separation majority voting requires

Section 4 assigns 0.100 to the selection loss and derives what a verifier must do to collect it. This section reports what nine scorers actually do, why combining them does not help, and why the gap between the requirement and the measurement is the stable fact of this project.

5.1 Setup

We score every sample with a function v, standardise within the problem to $z = (v - \bar{v})/\sigma_v$, and rank candidate answers by $s(a) = \log n_a + \gamma\, z(a)$. We evaluate a scorer by *within-problem pairwise AUC*: over pairs of one correct and one incorrect sample from the same problem, the fraction in which the correct sample scores higher. Chance is 0.5. This measure ignores between-problem difficulty, which is what we want, since the aggregator only ever compares samples inside a problem.

We report AUC twice: over all problems, and over the subset where plurality voting is already wrong. Only the second subset can change the answer. A scorer that is excellent overall and at chance on that subset is worth nothing.

5.2 Five scorers on one basis, none within a factor of three

We scored candidate answers nine ways: a trained outcome reward model (a LoRA head on the same base), four zero-shot prompted judges (plain, adversarial, critique-then-judge, and a reverse formulation that substitutes the candidate answer into the problem constraints), the base model's length-normalised log-probability, and the two fine-tuned generators of §8 used as scorers rather than as samplers.

Only five are measurable on the primary basis. The four prompted judges and the log-probability baseline were scored against a different generation file, at $k = 32$ on 300 problems, and their scores do not align sample-for-sample with the $k = 64$ file every other number in this paper uses. We report them separately below rather than in the same table, because putting incomparable bases in one column is the error §10 is about.

Three readings follow.

	adversarial	neutral	difference	
AUC, all problems	0.7456	0.7545	+0.0090	(2.0 SE)
AUC, plurality-wrong only	0.6240	0.6599	+0.0359	(3.2 SE)
Fused accuracy, $\gamma = 0.5$	0.8260	0.8320	+0.0060	(1.9 SE)

Table 6: Retraining the reward model with a neutral judging stance. The verifier improves on the subset that decides the answer; the answer does not change.

First, discriminative training buys real signal, and it buys it where the answer is decided: the reward model sits 5.6 standard errors above chance on the plurality-wrong column, against 1.8 for either fine-tuned generator. Second, the advantage carries across both columns — +0.15 overall and +0.12 on the hard subset, relative to the generators — so the shortfall is not an artefact of measuring on easy problems. The best column we have is the relevant one, and it is a factor of 3.8 short.

Third, and least expected, the last row. It is the same trained reward model, scoring the same candidates, under a judging prompt that does not match the one it was trained with. Its AUC is 0.4933: chance, within one standard error, on both columns. The discriminative ability does not survive a prompt change. Whatever the model learned is tied to the surface form it was trained on, not to the arithmetic. We return to this below, because it is the same effect that the stance retraining exploits.

The four prompted judges, on their own basis. On 300 problems at $k = 32$, the plain-stance judge reached an AUC of 0.681 and the adversarial-stance judge 0.670 on the full set; length-normalised log-probability reached 0.588 on the plurality-wrong subset, where the trained model reached 0.587. Those two numbers are the underpowered comparison discussed at the end of this section. No prompted judge exceeded the trained model on any basis we measured, and none is close to 0.940.

Changing the training prompt improves the verifier. The reward model above was trained with an adversarial judging stance. An early zero-shot comparison had found the adversarial stance worse than a neutral one (0.670 against 0.681), which implied the trained model had been given the weaker of the two. We retrained it with the neutral stance, holding data and every hyperparameter fixed, and scored with the matching stance.[1]

The score improved by 3.2 standard errors on exactly the problems where plurality voting fails. Fused accuracy moved by +0.0060 at 1.9 standard errors: above the run-to-run variation of 0.002, below the two-standard-error bar we set for every other claim in this paper. We adopt the retrained model, because its acceptance criteria were registered in advance and both were met.

We cannot say the verifier got better. The retraining changed the judging stance in the training data *and* in the scoring prompt, together, and the two are confounded. The last row of Table 5 is why that matters: the same weights scored under a mismatched prompt fall to chance. A model whose discriminative ability is that sensitive to surface form may gain 0.0359 from a better judging policy, or from nothing more than the training and scoring forms agreeing more closely. Separating the two needs the other diagonal of a 2×2 — each stance scored under both prompts — which we

[1]The two runs read the same source file and record 26,023 and 26,034 training rows respectively. We did not determine the cause of the 11-row difference, which is 0.04% of the data.

did not run. What we can report is the measurement: under a matched neutral stance the model ranks correct answers higher, and the answer it produces changes by 0.006.

This number is a correction. The judgement that adopted the retrained model recorded the fused accuracies as 0.8240 and 0.8260, a difference of +0.0020. Neither figure is reproduced by either answer parser on the same generation file; measured again with the parser this paper reports throughout, the pair is 0.8260 and 0.8320. The adoption decision is unaffected — it passes by a wider margin, not a narrower one — but the effect is three times the size we recorded, and every statement we made about it downstream was wrong in the direction that suited our thesis. We give the corrected number and the correction together for that reason.

An earlier, underpowered measurement pointed the other way. On a smaller basis (300 problems, $k = 32$, 59 plurality-wrong problems) the same trained model scored 0.587 ± 0.064 on the hard subset, statistically indistinguishable from untrained log-probability at 0.588. We recorded that as a finding and treated the verifier route as closed. The larger basis reverses the reading: on the 100 plurality-wrong problems where a correct sample exists, the standard error falls to 0.031 and the separation becomes visible. The earlier claim was a failure to detect, not a demonstration of absence. We report the correction in §10 together with two other instances of the same error.

5.3 The requirement, and the size of the shortfall

Signal is present. It is the wrong size.

Overturning a plurality-wrong problem requires the correct answer to overcome its vote deficit under $s(a) = \log n_a + \gamma z(a)$. Across plurality-wrong problems the median deficit corresponds to $\Delta z > 2.2\sigma$. Under a normal approximation, $\Delta z = \sqrt{2}\,\Phi^{-1}(\text{AUC})$, so an AUC of 0.624 is a mean separation of 0.447σ and 2.2σ requires an AUC of 0.940. An earlier draft of this analysis put the requirement at 0.85; that figure was an inversion error, and correcting it widens the shortfall rather than narrowing it.

This single comparison explains every negative result in this section. It also sets the scale of the fusion gain we do adopt. Against plurality voting alone, $\gamma = 0.5$ with the retrained verifier gains +0.0070 on validation (1.4 SE); with the original verifier it gained +0.0010 (0.2 SE). That is what a separation between 0.447σ and 0.583σ buys when it is allowed to break near-ties and nothing more.

6 The requirement is a statement about the verifier and the weight together

6.1 We wrote it as one

The threshold above is not a property of the task. Reading the fusion rule again,

$$\log n_c + \gamma z_c > \log n_p + \gamma z_p \quad \Longleftrightarrow \quad z_c - z_p > \frac{\log n_p - \log n_c}{\gamma}, \tag{3}$$

so the required separation scales as $1/\gamma$. Our $\Delta z > 2.2\sigma$ is the median log-count deficit of 1.10 divided by the $\gamma = 0.5$ we had already chosen. Stated as a bar the verifier fails to clear, it is circular: γ was set small *because* the verifier is weak, and the small γ was then used to derive how strong a verifier would have to be.

γ	required Δz	required AUC	measured 0.6599
0.5	2.200σ	0.9401	3.8× short
1.0	1.100σ	0.7817	1.9× short
2.0	0.550σ	0.6513	**met**
4.0	0.275σ	0.5771	**met**

Table 7: The analytic requirement as a function of the fusion weight. At $\gamma \geq 1.89$ the verifier we trained satisfies it.

The escape route the algebra opens, the measurement closes. Table 7 says that raising γ makes our verifier sufficient. We swept γ over twelve values and three ways of aggregating z within a candidate answer, on the primary generation file, and measured fused accuracy directly (Figure 5).

γ	0.5	2.0	3.0	6.0
analytic verdict	3.8× short	met	met	met
measured maj@64 (mean aggregation)	0.8320	0.8070	0.7910	0.7590

Table 8: Every γ at which the algebra calls the verifier sufficient scores below the configuration at which it calls it insufficient.

The whole region in which the requirement is satisfied lies below plurality voting alone. Raising γ lowers the bar and lowers the score at the same time, because a verifier that cannot rank correct answers first does not become useful by being trusted more. **The measured curve, not the threshold, is the object we should have reported.** We keep the derivation because it explains the shape of the curve, and we state its limitation here rather than in the limitations section, because a reader who takes 2.2σ as a task constant will draw the wrong conclusion from it.

What the fusion actually does at the operating point. At $\gamma = 0.5$ with mean aggregation, fusion overturns 17 problems that plurality voting gets wrong and breaks 10 that it gets right: net +7 of 1000. That is the whole of the effect, and it is the concrete form of "a 0.583σ signal breaking near-ties". The variance of the aggregate difference, $\mathrm{Var}(\bar{z}_c - \bar{z}_p) = 1/n_c + 1/n_p$, is why anything is overturned at all: on a problem with one correct sample against nine wrong ones the standard deviation is 1.05, and a mean shift of 0.583 clears a 2.2 threshold 6.3% of the time. The threshold is not deterministic, and treating it as such understates a weak verifier's value on exactly the problems where its value is realised.

The grid confirms the operating point, and shows why grids should not be trusted to choose it. Across twelve weights and three aggregation functions, the highest cell is 0.8320 and three cells reach it: mean aggregation at $\gamma = 0.5$ — the configuration already submitted — and maximum aggregation at $\gamma = 1.0$ and $\gamma = 1.5$. No setting beats the operating point.

The preregistered selection rule was to take the best cell on the first half of the validation set and evaluate it on the second. That rule chose maximum aggregation at $\gamma = 0.125$, which scored 0.8300 on the first half and lost 0.0120 against the operating point on the second. A 36-cell grid maximised on 500 problems selects noise. We report this because the same procedure, run without the held-out half, would have produced a confident recommendation to change the submitted system.

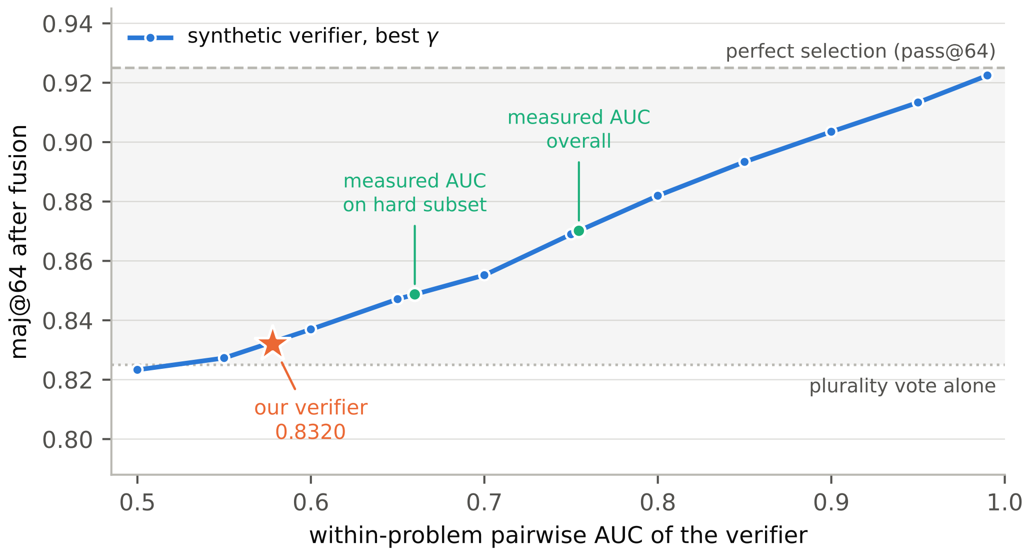

Figure 3: Fused accuracy against verifier quality, at the best weight for each quality. Green marks the two AUCs we measured for our reward model. The star is what it actually delivers. The horizontal distance between them is the subject of this subsection.

6.2 What a verifier of a given quality is actually worth

The analytic requirement rests on a normal approximation and a median. Both can be removed. We injected a synthetic verifier of controlled quality into the real generation file — correct samples drawn from $\mathcal{N}(\mu, 1)$ and incorrect ones from $\mathcal{N}(0, 1)$, with $\mu = \sqrt{2}\,\Phi^{-1}(\text{AUC})$ — and applied the real fusion rule to the real candidate sets and real vote counts. Forty draws per cell, eleven quality levels, five weights, and the same 1,000 problems. Nothing is assumed except that the verifier's errors are independent of the generator's.

Verifier AUC	0.55	0.65	0.75	0.85	0.95
maj@64, best γ	0.8273	0.8471	0.8626	0.8933	0.9134
best γ	0.5	1.0	1.5	2.0	2.0
selection loss recovered	2%	22%	38%	68%	88%

Table 9: A synthetic verifier of known quality, measured on the real generation file. Recovering half the selection loss takes an AUC of 0.773; three quarters takes 0.883. Full recovery is not reached even at 0.99, because at that point the fusion weight that would collect the rest also destroys correct pluralities.

Our verifier behaves like a much worse one than it measures. Our reward model scores an AUC of 0.7545 over all samples and 0.6599 on plurality-wrong problems. A synthetic verifier at 0.7545 delivers 0.8712 and one at 0.6599 delivers 0.8495. Ours delivers 0.8320. Inverting the curve, the homogeneous verifier that would produce 0.8320 has an AUC of **0.578**.

Three deficiencies stack, and the simulation separates them:

	AUC	selection loss recovered
measured, all samples	0.7545	46%
$\hookrightarrow$ weaker where the answer is decided	0.6599	24%
$\hookrightarrow$ errors that coincide with the generator's	0.578 (effective)	7%

The first step was already visible in Table 5. The second was not. A synthetic verifier's mistakes are independent of the generator's by construction; a reward model trained by adapting the generator's own weights makes its mistakes on the same problems, so the samples it ranks highest on a hard problem are the ones the generator was already most confident about — and most confidently wrong about. **Sharing a base costs more than being weak.**

An independent check on the effective AUC. The simulation says the best weight for a homogeneous verifier at AUC 0.578 is $\gamma \approx 0.75$, and for one at 0.7545 it is $\gamma \approx 1.5$. Our grid search over twelve weights and three aggregations, run on measured scores with no knowledge of any of this, chose $\gamma = 0.5$. The operating point we arrived at empirically is the one that suits a 0.578 verifier, not a 0.7545 one. Two independent routes give the same answer about what our verifier is worth.

6.3 Combination does not close the gap, and theory overstates how much it helps

Signals with imperfect correlation should combine to something stronger than either. Under the standard result for combining forecasts,

$$\mathrm{IC}_{\mathrm{comb}} = \sqrt{\frac{\mathrm{IC}_1^2 + \mathrm{IC}_2^2 - 2\rho\,\mathrm{IC}_1\mathrm{IC}_2}{1-\rho^2}}, \qquad \mathrm{IC} \approx 2\,\mathrm{AUC} - 1. \tag{4}$$

We measured the inputs and the output. Correlations between the trained reward model and the two fine-tuned generators are 0.182 and 0.195 — low, which the formula rewards. Predicted gains are +0.0147 and +0.0166. Measured gains, with the weight chosen by grid search on the full set, are +0.0018 and +0.0025: smaller by factors of 8.2 and 6.6. An earlier three-way combination behaved the same way.

The subsection above identifies the mechanism. The formula treats a global correlation of 0.18 as if the two signals erred independently. They do not: both are built on the same checkpoint, and the simulation shows that our verifier collects 7% of the selection loss where a signal of independent errors and the same measured AUC would collect 46%. Errors that coincide on the hard subset are invisible to a global correlation and fatal to a combination. The operational conclusion follows: **low correlation is necessary for a combination gain, not sufficient, and measuring it globally will not tell you which case you are in.**

6.4 A genuinely independent second signal, and it still does not pay

Every combination above pairs the reward model with something that shares its base and its training history. The generation files carry one signal that does not: the base model's mean token log-probability for each sample, recorded at generation time at no extra cost. Within problems it correlates with the reward model at a median of 0.179, against 0.974 between the two fine-tuned adapters. If low correlation were the binding constraint, this is the signal that should pay.

We scored it under $s(a) = \log n_a + 0.5\, \bar{z}_{\mathrm{ORM}}(a) + \gamma_\ell\, \bar{z}_\ell(a)$, holding the reward-model weight at its operating value so that only one degree of freedom was searched.

γ_ℓ	full set	first half	second half
0.00	0.8320	0.8280	0.8360
0.25	0.8340	0.8300	0.8380
0.50	0.8370	0.8320	0.8420
0.75	0.8370	0.8300	0.8440
1.00	0.8300	0.8260	0.8340
2.00	0.8000	0.7940	0.8060

Table 10: Adding the log-probability signal to the fusion. The curve is smooth and single-peaked, unlike the grid in §9 whose winner was a spike.

Maximised on the full set the combination gains +0.0050. That figure is not a test, and we ran two.

The first selected γ_ℓ on the first half and evaluated on the second: +0.0060 at 1.8 standard errors, missing the two-standard-error clause. That gate was unpassable, for the reason §10 describes. The second, run after the first and reported alongside it rather than in place of it, used two-fold cross-validation for power: +0.0040 at 1.1 standard errors. The two folds selected different weights, 0.50 and 0.75, which is what an unstable optimum looks like. Both trials miss. We do not adopt the signal.

One measurement remains. We submitted the combination to the public leaderboard once, having recorded in advance that the result would not change the configuration and that we expected no movement. It scored 0.79783 against 0.79663: one problem in 831.

So the reward model, two fine-tuned generators, and an independent confidence signal all combine to within a problem of the reward model alone. The conclusion of the previous subsection survives a second, harder test. **Low correlation buys nothing when neither signal is large enough to overturn a vote.**

6.5 What the correlation structure does show

The two fine-tuned generators correlate with each other at 0.974. They were produced by different objectives — supervised distillation from a stronger teacher in one case, preference optimisation on self-generated pairs in the other — and they score candidate solutions almost identically. Taken with §8, where the same two runs move validation accuracy by indistinguishable amounts, this suggests the limit we keep meeting belongs to the base model rather than to any particular training procedure.

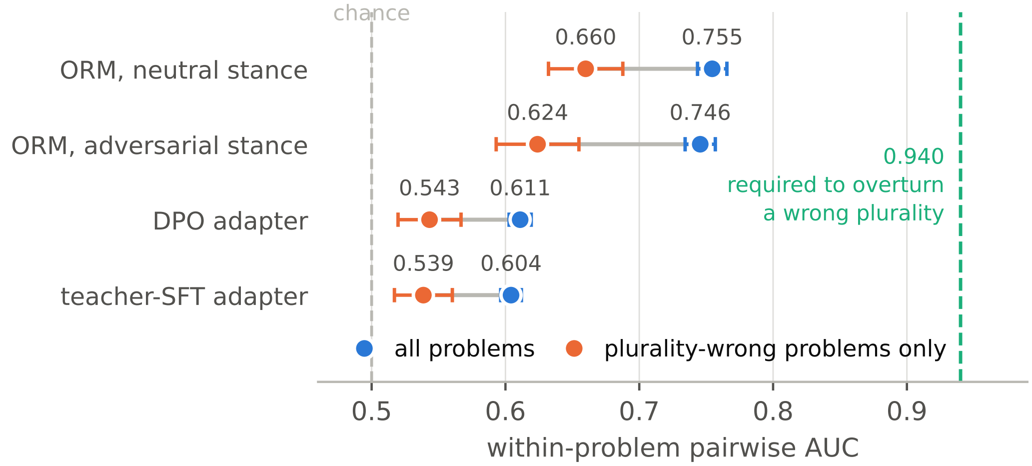

Figure 4: Four scorers, measured on 1,000 problems (blue) and on the 100 plurality-wrong problems a selector could still recover (orange). All four sit far left of the separation the fusion rule requires. Bars are ± 1 standard error from a paired bootstrap over problems, 2000 resamples.

7 Coverage rises by 0.006 to 0.014 and accuracy does not follow, with or without a verifier

Section 4 sets a budget of 0.075 for anything that enlarges the hypothesis space. We spent it four ways. All four raised coverage or left it unchanged; none of them moved selection accuracy. Figure 6 places every intervention against the line of perfect transfer.

7.1 Decoding: temperature and sample count

Raising the sampling temperature from 0.8 to 1.15 left pass@64 at 0.930, indistinguishable from the 0.8 setting. Pooling 64 samples at each temperature reached pass = 0.935, and plurality accuracy fell below both single-temperature settings. Doubling k from 64 to 128 within one run added 0.000 to maj@k, measured as a paired comparison on subsamples of a single generation file.

Checkpoint pooling, the fourth entry in Table 7.4, pools the 64-sample generations of two LoRA checkpoints of the same base. It was run on the retired 300-problem basis before we fixed the primary one, and we did not compute intervals for it; we report it as a range across two checkpoint pairs and give it no weight in the argument.

We then repeated the entire generation once under the identical configuration and pooled the two runs, which gives 128 samples with no change of model, prompt, or temperature. Coverage rose by 0.0070 (2.7 SE, bootstrap) and plurality accuracy by 0.0020 (1.0 SE). The two conditions differ in one respect, the number of samples, and the coverage gain does not reach the answer.

7.2 Prompt diversity

Temperature perturbs samples inside one conditional distribution $p(\cdot \mid q, \text{prompt})$. Changing the prompt changes the conditioning, so the reachable set of hypotheses can differ. We generated 64

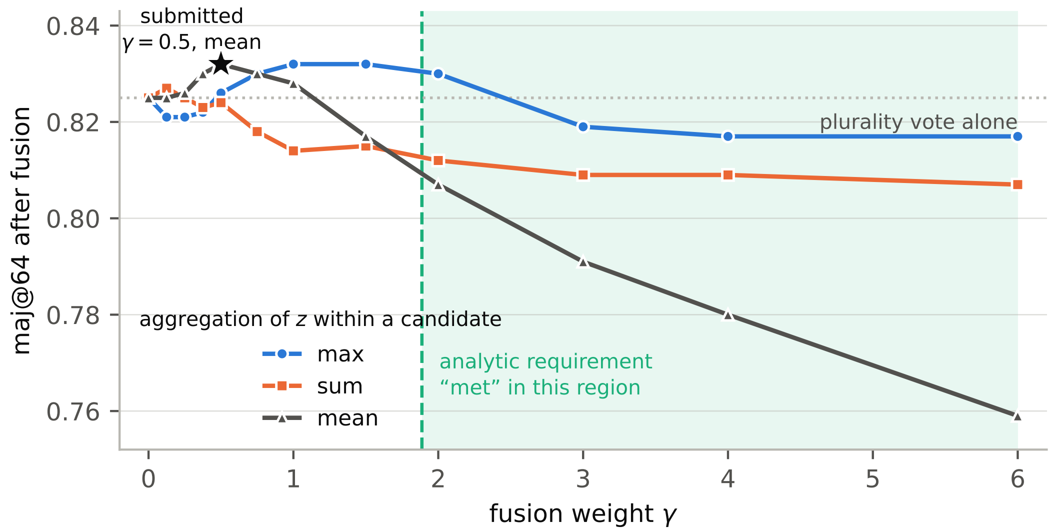

Figure 5: Fused accuracy against the fusion weight, for three ways of aggregating the verifier score within a candidate answer. The shaded region is where the analytic requirement of §4 is satisfied by the verifier we trained. All of it lies below plurality voting alone.

samples under a second prompt whose instructions differ in wording but not in content, and pooled them with the 64 samples from the prompt used throughout.

Coverage rose from 0.9240 to 0.9370, a gain of 0.0130 (3.7 SE). Plurality accuracy moved by −0.0040 (1.1 SE).

That 0.0130 is not the effect of prompt diversity. It compares 64 samples against 128, and pooling any second source of 64 raises pass@k by construction. The budget-matched control is the condition above: pooling two runs of the *same* prompt, also 128 samples, gains 0.0070. The prompt-diversity effect net of sample count is +0.0060, not +0.0130. We report both because the unpaired figure is the one a reader would otherwise carry away, and it is inflated by a factor of two. The conclusion is unchanged: both numbers reach coverage and neither reaches the answer.

We registered a dilution control before running: the second prompt had to score within 0.020 of the first on its own, so that a flat plurality result could not be explained by a weaker second source. It scored −0.0130, inside the bound. The transfer failure is therefore not attributable to quality dilution.

7.3 Why the gains do not transfer

The mechanism follows from how the newly reached answers appear. A problem that enters the covered set does so because one or two samples out of 128 produce the correct answer. That is not enough to change the mode. Fusion could in principle promote it, but promotion requires the separation computed in §4, which the verifier does not supply.

The result is consistent across four interventions with different mechanisms: pooling checkpoints, pooling temperatures, pooling prompts, and pooling repeated runs. In every case coverage moved and selection did not. We read this as empirical support, on a small model under a fixed-base constraint, for the *modal ceiling* described by prior work on test-time scaling: without a verifier

Axis	Intervention	Basis	target metric	accuracy
generation	checkpoint pooling	earlier	pass +0.007 to +0.013	0.000
generation	temperature pooling	earlier	pass +0.010	negative
generation	prompt pooling	primary	pass +0.0060 (matched)	−0.0040 (1.1 SE)
generation	repeated-run pooling	primary	pass +0.0070 (2.7 SE)	+0.0020 (1.0 SE)
selection	verifier stance	primary	AUC +0.0359 (3.2 SE)	+0.0060 (1.9 SE)
selection	second scoring signal	primary	— (independent, $\rho = 0.18$)	+0.0040 (1.1 SE)
both	pooling *with* verifier	primary	pass +0.0140	−0.0030 (0.7 SE)

Table 11: Seven interventions across both axes. "Primary" is 1,000 problems at $k = 64$ with paired bootstrap intervals; "earlier" is the retired 300-problem, $k = 32$ basis, on which we did not compute intervals. Four of the five leave submitted accuracy inside the run-to-run variation of 0.002. The largest, the verifier retraining, reaches +0.0060 at 1.9 standard errors. None clears two.

that can identify correctness, selection saturates at the rate at which the correct answer is also the most frequent one, independently of the sample budget.

7.4 Both axes fail the same way

The four interventions above raised coverage without raising accuracy. The verifier retraining in §5 is the same shape of result on the other axis — a measured improvement in the quantity the design targets — and it is the one case where the score moved with it, by +0.0060 at 1.9 standard errors. That is above the run-to-run variation of 0.002 and below the two-standard-error bar. We do not claim it as transfer and we do not claim it as null. It is also the one intervention whose mechanism we could not isolate: training stance and scoring stance moved together, and §5 shows the same weights falling to chance under a mismatched prompt. The closest thing to a success in this project is also the thing we understand least.

The requirement explains the shape. Expressed as a separation, the adversarial model delivers 0.447σ and the neutral model 0.583σ, against 2.2σ at the operating γ. Retraining closed the gap from a factor of 4.9 to a factor of 3.8, and bought 0.006 of accuracy for it.

7.5 Coverage still does not convert when a verifier does the choosing

Every result above selects by plurality with a verifier weight applied to the pooled candidates of a single source. The obvious objection is that we never gave the verifier the enlarged candidate set to choose from. We did, and it does not help.

We scored two additional 64-sample generations of the same 1,000 problems with the same reward model — one from a second prompt, one a verbatim repeat of the first configuration — and fused each pooled 128-sample set under the operating rule.

The ordering is the result. Coverage rises monotonically across the three rows, by 0.0070 and then 0.0140. Fused accuracy does not: it moves +0.0010 and then −0.0030, and the row with the most correct answers available is the row that gets the fewest of them. The net effect of prompt diversity over the budget-matched control is −0.0040 (0.9 SE).

The mechanism is the one §4 prices. A problem enters the covered set when one or two of 128 samples produce the correct answer. Doubling the pool also doubles the distinct wrong answers

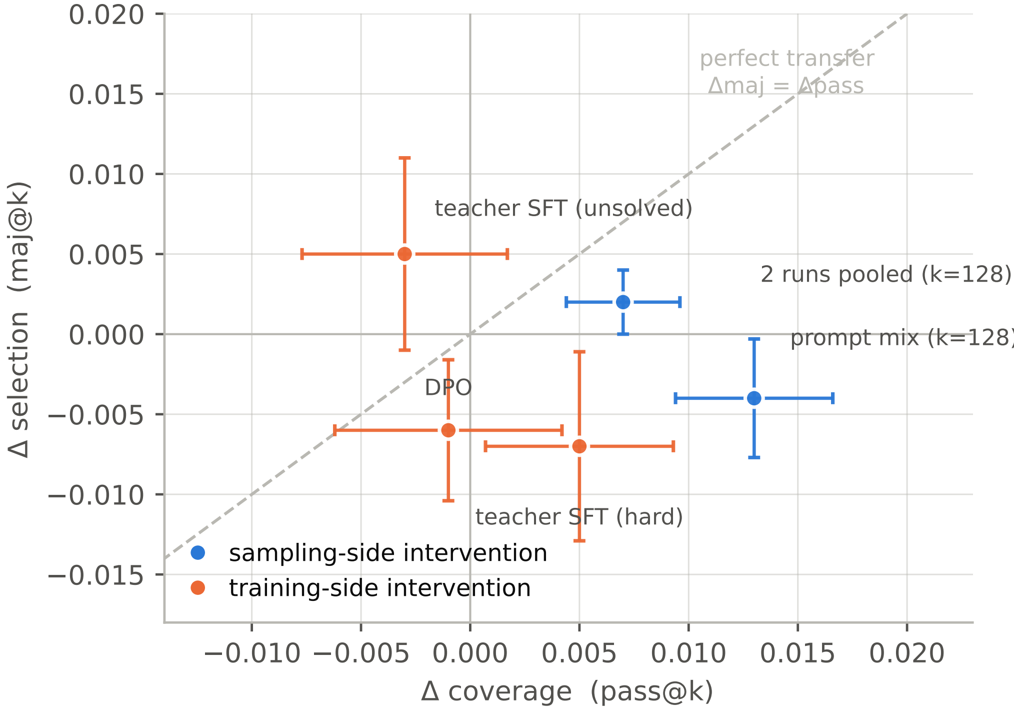

Figure 6: Every intervention moved its target metric and none moved the submitted answer. Intervals are paired bootstrap over problems, 2000 resamples; the shaded band is the run-to-run variation of 0.002 measured in §9.

competing for the plurality, so the vote deficit the verifier must overturn grows at the same time as the answer becomes available. A selector at 0.583σ against a requirement of 2.2σ cannot pay for the additional deficit, and the additional candidates dilute the mode. **Coverage is not a resource a weak selector can spend.**

8 Seven training runs, none above the base model

The competition fixes the base model and forbids merging any other model's weights, so the only training available to us is a LoRA adapter (Hu et al., 2022) on that base. We used it seven times, with four different sources of supervision. None of the seven improved the system, and the way they fail is more informative than the fact that they do.

8.1 What we tried

The external corpora removed capability rather than adding it. Training on twenty thousand public math solutions cost 12.4 points of sample accuracy at lr 10^{-4}, against an expected distribution-mismatch penalty of a few points. That figure comes from the retired 300-problem basis and was never re-measured on the primary file; we report it as an order of magnitude, not as a precise loss.

Candidate set	samples	pass	fused accuracy
prompt C only	64	0.9250	0.8320
prompt C + repeat of prompt C	128	0.9320	0.8330 (+0.0010, 0.4 SE)
prompt C + prompt B	128	0.9390	0.8290 (−0.0030, 0.7 SE)

Table 12: Pooling with the verifier in the loop. The condition with the highest coverage has the lowest accuracy.

Supervision	Source	Rows	Outcome
Self-distillation	the model's own correct solutions	14,254	no gain
External corpus 1	`OpenMathInstruct-2`	20,000	sample acc. −0.124
External corpus 2	`orca-math-200k`, lr 10^{-4}	20,000	−0.079
External corpus 3	`orca-math-200k`, lr 2×10^{-5}	20,000	−0.055
Teacher distillation 1	stronger model, hard bucket	3,206	gate missed
Teacher distillation 2	stronger model, unsolved bucket	1,812	gate missed
Preference optimisation	DPO on self-generated pairs[2]	5,610	gate missed

Table 13: Seven adapter runs. The first four were evaluated on an earlier, smaller basis (300 problems, $k = 32$); the last three on the primary basis (1000 problems, $k = 64$) with paired bootstrap intervals. An eighth run was aborted mid-generation and is excluded.

8.2 The last three runs, measured the same way

The three runs below share a basis, so they can be compared directly against the base model on the identical generation file, with paired bootstrap intervals over problems.

Sample accuracy falls in all three, each time by at least 1.9 standard errors. The data differ (teacher solutions for hard problems, teacher solutions for unsolved problems, self-generated preference pairs), the objective differs (supervised likelihood versus preference), and the size differs by a factor of three. The direction does not.

Plurality accuracy, in contrast, moves within noise and changes sign across the three: −0.0070, −0.0060, +0.0050, none above 1.4 standard errors. Coverage likewise. The one consistent effect is a small loss of solver quality, and it does not reach the answer either — the same failure to transfer that §7 reports for interventions that pointed the other way.

8.3 Two claims we do not make

We do not claim a mechanism. Three candidate explanations appear in the literature and in our own earlier notes: diversity collapse, length compression, and catastrophic forgetting. None fits all three runs. Coverage rose in the run we had attributed to diversity collapse. Output length fell meaningfully in exactly one run, and that run is also the one where plurality accuracy rose. We report the pattern and leave the mechanism open.

We do not claim that distillation cannot work here. The teacher solved only 26% of the unsolved problems with a repeat, which left 409 problems against a planned 851. That arm was registered with an asymmetric reading rule for this reason, and it is reported as not demonstrated

	Δ sample acc.	Δ pass@64	Δ maj@64
Teacher distillation (hard)	−0.0038 (2.1 SE)	+0.0050 (1.2 SE)	−0.0070 (1.2 SE)
Preference optimisation (DPO)	−0.0028 (1.9 SE)	−0.0010 (0.2 SE)	−0.0060 (1.4 SE)
Teacher distillation (unsolved)	−0.0059 (3.3 SE)	−0.0030 (0.6 SE)	+0.0050 (0.8 SE)

Table 14: Change relative to the untrained base on the same generation file. Intervals are paired bootstrap over problems. Run-to-run variation is 0.002 (§9); the sample-accuracy column clears it in all three runs, and the two selection columns do not clear it in any.

rather than rejected. Similarly, the preference-optimisation run used a single conservative setting ($\mathrm{lr} = 5\times10^{-6}$, $\beta = 0.1$, one epoch); pushing harder would move the distribution further and would also reintroduce the risk the conservative setting avoids. Exploring that trade-off is more than a one-shot arm can carry.

8.4 One observation that survives all of it

Used as scorers rather than as samplers, the distillation adapter and the DPO adapter correlate at 0.974 (§5). Two different objectives, two different data sources, and nearly the same ranking function. Together with the table above, this points at the base model rather than at any procedure: what we could reach with a low-rank adapter on this checkpoint, under the data we could legally obtain, was already close to what the checkpoint does on its own.

9 Rerunning one configuration changes 42 answers, 0.002 accuracy, and one leaderboard rank

Effects in this paper are between 0.003 and 0.013. At that size the measurement apparatus is part of the result, so we report it in full: what varies, by how much, and which comparisons the variation invalidates.

9.1 Generation is not deterministic

Running the identical configuration twice, with the same seed, produces different text for every problem. The inference engine batches requests dynamically, and the seed fixes sampling but not the composition of batches, so floating-point accumulation order differs. In two full repeats:

Repeat	problems	answers changed	aggregate accuracy change
Leaderboard, same machine	831	14 (1.7%)	0.79783 → 0.79663 (0.0012)
Validation, same machine	1000	42 (4.2%)	maj@64 0.8250 → 0.8230 (0.0020)
Leaderboard, rebuilt machine	831	18 (2.2%)	0.79663 → 0.79783 (0.0012)

Individual answers churn at a few percent. Aggregate accuracy moves by 0.002, because the flips run in both directions and cancel. We state the reproducibility claim from the direct measurement rather than from the flip count: **accuracy reproduces within** ±0.004. An earlier estimate of ±0.016, derived by treating flips as independent, was an overestimate; we report the correction because the two numbers lead to different conclusions about the rest of the paper.

Rebuilding the machine. The third row is a different experiment. We destroyed the rented instance and rented a new one, restored code, weights and data from the release archive, and re-ran the leaderboard configuration. The library set was pinned and identical except for the CUDA build of PyTorch, which the new image supplied as 13.0 against 12.8. Eighteen answers changed and the score moved by one problem, in the opposite direction to the same-machine repeat.

Where the changes fall is more informative than how many there are. Ranking each problem by the vote margin between its top two answers, the median margin over all 831 problems is 59; over the eighteen that changed it is 2, and none exceeds 16. Instability is confined to the near-ties, which is what the probabilistic threshold of §6 predicts: where $n_c \approx n_p$, an arbitrarily small perturbation of $\bar{z}$ moves the decision.

Comparing the fused submissions against their plurality-only controls separates the two sources. Of the eighteen changes, thirteen are ones where the plurality answer itself changed and the fusion followed; five are ones where the plurality answer held and the verifier moved the result. In the other direction, the plurality answer changed on twenty-one problems, and on eight of them the fused answer held. The verifier absorbs more instability than it introduces — eight against five — but with thirteen discordant problems that is not evidence (McNemar exact, two-sided $p = 0.581$), and we record it as a direction, not a result. It does not revive the variance-damper hypothesis of §10.

Both counts trace to one cause — the generated text differs — and they separate by where the difference surfaces. On thirteen problems the vote moved and the verifier followed it; on five the vote held and the verifier's response to the changed text moved the answer. Neither figure isolates the scoring pass, which sees a different input in every case. Whether scoring is itself sensitive to the rebuild is a separate question, answered by re-scoring a fixed generation file on both machines.

9.2 Which comparisons this invalidates

- **Within-file comparisons are unaffected.** Fusion gain, the γ sweep, the k ablation, and the regime analysis all draw subsamples from a single generation file. Nondeterminism does not enter.

- **Across-run comparisons of small effects are not usable.** We had recorded a claim about the transfer of a k increase to the leaderboard based on two separate runs. Run-to-run noise exceeds the effect. We withdrew the claim and replaced it with "not measurable"; the underlying conclusion was re-derived from a within-file comparison and stands.

- **Most reported effects do not clear the bar, and we say so.** Of the seven interventions in Table 7.4, six leave submitted accuracy inside ± 0.002 and the seventh reaches 0.006 at 1.9 standard errors. An earlier draft of this section claimed every reported effect exceeded three times the run-to-run variation. That was false, and it was false in the direction that made our measurements look stronger than they are. Sampling intervals throughout come from a paired bootstrap over problems with 2000 resamples; they do not include the generation-nondeterminism component, so a "3.7 SE" on an across-run comparison overstates significance by an amount we cannot quantify.

9.3 We searched more configurations than we report as tests

Preregistration fixes the gate for each arm. It does not correct for the number of arms. Across the project we evaluated nine scorers on two subsets, seven interventions, twelve fusion weights crossed with three aggregation functions, and nine weights for a second scoring signal — on one validation set of 1,000 problems. Naive standard errors on the winner of a grid are not the standard errors of a prespecified test.

Two devices keep this from contaminating the conclusions. Every grid result reported as a decision was evaluated on a held-out half that played no part in selecting it, and both times the held-out half rejected the grid winner: a 36-cell sweep chose a cell that then lost 0.0120, and a 9-cell sweep chose a weight whose cross-validated gain fell to 1.1 standard errors. And the conclusions this paper draws are nulls. A multiplicity correction widens intervals, which makes a null easier to sustain, not harder. Where we report a positive effect — §5's +0.0060 — we state that it does not clear two standard errors and we do not build on it.

9.4 What a null result here excludes

§10 commits us to stating the effect size a null excludes. On the primary basis, a paired comparison over 1,000 problems has a standard error near 0.005 for accuracy differences, so the smallest effect a two-standard-error gate can detect is about 0.010; on a held-out half of 500 problems it is about 0.0068. Our nulls therefore exclude effects larger than roughly one percentage point on the full set. They do not exclude effects of 0.002 to 0.006, and two of our own gates were set inside that undetectable band — an error we describe in §10.

9.5 A rank change from re-running the same configuration

We submitted the repeat of our own leaderboard configuration as a measurement, not as an entry: the same model, settings, and seed, generated a second time. The score moved from 0.79783 to 0.79663, one problem out of 831.

At the time, the fourth and fifth positions on the public leaderboard were separated by 0.0012, and our repeat landed on the fifth-place score exactly. Positions two through six spanned 0.0096, within the range that a single re-run of one system can traverse. Only the leading entry, 0.0229 above us, sits outside it.

We note this because leaderboard-adjacent work rarely reports run-to-run variance, and the ordering of closely-spaced entries may carry less information than it appears to. We make no claim about other participants' systems, whose variance we cannot observe.

9.6 Compute

All experiments ran on a single rented A100 SXM4 40GB. Generation throughput was approximately 10^4 tokens per second; one validation pass (1000 problems, $k = 64$, 3072 max tokens) takes about 50 minutes, and the final test pass ($\approx$ 2000 problems) about 2.5 hours. Adapter training took 50 minutes to 1 hour 50 minutes per run depending on data size. Total GPU time for the project was approximately 40 hours. External API calls for teacher data cost about $22 and used no local compute.

Two Python environments are required, because the inference and training stacks pin incompatible versions of the same library; Appendix A gives both.

9.7 Determinism where it exists

Adapter merging, answer extraction, tie-breaking in the vote, and the fusion score are all deterministic. Given a fixed generation file, every downstream number in this paper reproduces exactly. Nondeterminism is confined to the generation step, which is the property a reproduction attempt should be told about in advance.

10 Preregistration stopped three decisions and missed three underpowered claims

Every experiment in this paper was registered before it ran. A registration names the hypothesis, the acceptance gate, the number of runs allowed, and the rule for reading the outcome. The record is a single append-only file, timestamped, holding 47 numbered entries from the first day to the last. We describe the protocol, what it prevented, and — more usefully — the three things it failed to prevent.

10.1 The protocol

A registration fixes four items.

Item	Content	Purpose
Gate	numeric threshold on a named metric	fixes the bar before the number exists
Trials	one, in every arm but two	a gate that can be retried is not a gate
Reading rule	miss $\Rightarrow$ reject (6 of 8 arms)	forbids reinterpreting a miss
	miss $\Rightarrow$ not demonstrated (2 arms)	forbids claiming absence from low power
Prediction	the author's stated expectation	makes a wrong call as visible as a right one

Table 15: The four items every registration in this paper fixes before its measurement runs.

The append-only ordering is the mechanism, not the file format. Chronology is what makes "registered before observation" checkable by a reader.

10.2 What it prevented

Three attempts were stopped by the protocol rather than by judgement. We proposed tuning the fusion weight γ against the public leaderboard, which would have selected a hyperparameter on the evaluation signal; the registration required the sweep to happen inside validation. We proposed submitting a configuration that had missed its gate, on the reasoning that validation and leaderboard sometimes disagree; the one-shot rule forbade it. And a training set that we had described as targeting hard problems turned out to be 97% a different bucket, which the registration surfaced because the composition had been written down in advance.

Original claim	Basis	Larger-sample reading
Teacher yield on unsolved items is 71.2%	$n = 312$	33.0% at $n = 1595$; also confounded
Teacher SFT collapsed sample diversity	no SE reported	coverage rose; effect within noise
No scorer beats chance where voting fails	59 problems	AUC 0.624, 4 SE above chance

Table 16: Three claims recorded as findings and later withdrawn. Each came from a small sample, and in two cases no standard error was reported alongside the claim.

10.3 What it did not prevent: three underpowered claims

The protocol governs *when* a decision is made. It says nothing about whether the measurement can support the decision. Three times we recorded a null result as a finding, and three times a larger sample reversed it.

The common failure is not the sample size as such. It is that a null result was written as a statement about the world ("nothing works here") when the data only supported a statement about the measurement ("we could not detect an effect of this size"). We now write negative results in the second form, and we state the effect size the result excludes.

10.4 A gate that the baseline itself could not pass

One error was in a threshold, and it propagated to every training judgement. Our training gate required pass@64 ≥ 0.9367, a figure taken from an early measurement of the base model. Measured on the generation file that all later judgements used, the base model scores 0.9240. The gate was therefore unreachable by the base model itself, and every training arm failed that clause regardless of its merit.

We made the same error a second time, on the last day, after having written the first one up. Testing whether a second scoring signal improves the fusion, we set a gate at two standard errors on a held-out half of 500 problems. The standard error there is 0.0034, so the smallest detectable effect was 0.0068. The effect under test was 0.0060. **The gate could not have passed whatever the signal was worth.** It returned +0.0060 at 1.8 standard errors and we recorded a miss, which is the correct reading of that gate and an uninformative one about the hypothesis.

The pattern is the same both times: a threshold chosen from a plausible-sounding quantity — an early measurement in one case, a conventional significance bar in the other — without checking it against the resolution of the measurement that would have to clear it. A gate should be validated before it is used, by asking what the instrument can detect. We now do that, and we report both failures because the second one happened after we understood the first.

The consequence was not only mechanical. One arm was recorded as failing because "coverage collapsed". Measured against the base model on the same file, its coverage had *risen* by 0.0050; what fell was sample accuracy, by 0.0038. The stated mechanism was the reverse of the observed one, and it survived for five days because the unreachable threshold produced a miss that looked like confirmation.

We found this while building a script to extract numbers for a talk, when its output disagreed with the gate script. The discrepancy was a parser difference; the threshold problem was underneath it. **An unrelated task did the auditing that the protocol did not.**

What we changed, and what we did not. We did not move the threshold. Adjusting a preregistered criterion after seeing results removes the property that made it worth registering. The judgements stand as recorded: all training arms missed. What we added is disclosure — the threshold and the evaluation set come from different generation runs, and the clause is not passable — plus a second reported layer, a paired comparison against the base model on the identical file, with paired bootstrap intervals. The affected mechanism claim was downgraded to "not established".

10.5 The prediction ledger

Registrations carry an author prediction, which makes our own reliability measurable. Across the project the split is clean: **eight of eleven predictions about magnitude held; zero of five predictions about mechanism did.**

Registered prediction	Outcome	Class
repeat submission lands inside the reproducibility band	held	magnitude
final training arm moves accuracy by less than 0.005	held	magnitude
pooling raises coverage and not accuracy	held	magnitude
length collapse causes the distillation failure	refuted	mechanism
best-of-n helps most in the lowest-accuracy regime	refuted	mechanism
label noise dominates the unsolved bucket	refuted	mechanism
the verifier acts as a variance damper	refuted	mechanism
retraining the verifier stance misses both gates	refuted	mechanism
leaderboard fusion gain lands in $[0.000, 0.012]$	held ($\leq +0.012$)	magnitude
the retrained verifier does not transfer to the leaderboard	held (-0.0012)	magnitude
an earlier cross-run estimate of +0.012 is inflated	refuted (+0.0120)	magnitude
pooling with the verifier in the loop still fails	held	magnitude
a second scoring signal clears its gate	refuted	magnitude
relaxing the label-error filter yields 5–15× more candidates	refuted (52×)	magnitude

Table 17: Fourteen registered predictions. Eight of eleven magnitude calls held; zero of five mechanism calls did. The two magnitude misses are the last two rows, both made on the final day, and both concern the size of something we had not measured before.

We report this because it changes how the rest of the paper should be read. Where we give a number with an interval, we have a record of such statements holding. Where we explain *why* something happened, we have a record of being wrong, and we mark those passages as hypotheses.

11 Limitations: 3B, 64 samples, one problem distribution

One model, one task family. Everything here is measured on a single 3B checkpoint solving integer-answer word problems. The decomposition transfers to any setting with a sampling stage and a selection stage, but the numbers do not. Whether the selection loss dominates at other scales, or on tasks where errors concentrate differently, is untested.

The one positive result is confounded. The stance retraining changed the judging prompt in training and in scoring at the same time. §5 reports the measurement and not a mechanism, because the 2×2 that would separate a better judging policy from closer form-matching was not run. Anything built on that +0.0060 inherits the ambiguity.

The verifier space we could search was bounded by the competition rules. The rules fix the base model and forbid loading weights from any other. A process-supervised reward model — the standard next instrument after an outcome-supervised one — is therefore out of reach as an off-the-shelf component, and training our own would need step-level labels the benchmark does not carry. Our nine scorers are nine ways of using one 3B checkpoint. A reader should not read "no verifier was strong enough" as a claim about verifiers in general.

Four of nine scorers were measured on a retired basis. The prompted judges and the log-probability baseline were scored against a 300-problem, $k = 32$ generation and never brought onto the primary file. We report them separately for that reason and exclude them from the requirement comparison. Bringing them across is a few GPU-hours of re-scoring that we did not spend.

The verifier requirement is tied to our aggregator. The figure $\Delta z > 2.2\sigma$ is derived under $s(a) = \log n_a + \gamma z(a)$. A different combination rule — weighting by rank, by calibrated probability, or by a learned aggregator — would move that requirement. We did not sweep aggregator families, so the requirement should be read as specific to this rule rather than as a property of the task.

Two arms are underpowered. The unsolved-bucket distillation arm ran on 409 problems against a planned 851, because the teacher solved only a quarter of the bucket. The preference-optimisation arm used one conservative hyperparameter setting. Both were registered with reading rules that forbid us from treating a miss as a rejection, and we hold to that: those hypotheses are untested, not refuted.

Mechanism claims are weaker than magnitude claims. Our own record shows five of five mechanism predictions refuted and eight of eleven magnitude predictions held (§10). Where this paper offers an explanation rather than a measurement, it should be read at that reliability.

Run-level variance rests on two repeats. The ± 0.004 reproducibility figure comes from two independent repeats, one on each evaluation set. Two observations give a poor estimate of spread. The direction of the correction — that the flip-count estimate was too large — is solid; the exact bound is not.

Label noise remains in the benchmark. We removed the 627 problems the organisers identified, and Table 3 shows the removal was incomplete. A further set they acknowledged was never released. Our validation numbers therefore include an unknown quantity of unrecoverable problems, which bounds pass@k below 1.0 by an amount we cannot separate from genuine model failure.

The uncovered set is characterised from 30 of 75 problems. Table 3 rests on a sample of 30, classified twice. The defective-item rate carries a Wilson interval of $[9.5\%, 37.3\%]$, wide enough that the corrected ceiling of 0.9400 should be read as an estimate. The four wrong gold answers are certain — each is a computation — but the category boundaries are not: $\kappa = 0.68$ means roughly a fifth of the assignments would move under a third annotator, and we know the direction they move in, because both times the disagreement went the same way. If anything the rate we report is still low.

A Reproduction

A.1 Hardware and environment

GPU	NVIDIA A100-SXM4-40GB, driver 580.159.03
CPU	AMD EPYC 7542, 128 cores
RAM	2015 GB
OS	Ubuntu 24.04.4 LTS, kernel 6.17.0
Python	3.12.13

Two environments are required. Inference uses `vllm` 0.27.1 with `torch` 2.13.0+cu130 and `transformers` 5.15.1. Training uses `peft` 0.20.0 and `trl` 1.10.0 on the same torch build. The two stacks pin incompatible versions of `transformers`, so they cannot share a virtual environment. A complete `pip freeze` accompanies the code release.

The base model is `Qwen/Qwen2.5-3B-Instruct`. We record the SHA-256 of both weight shards with the release so that a reproduction can confirm it started from the same checkpoint.

A.2 Configuration of the submitted system

Base	`Qwen2.5-3B-Instruct`, weights unmodified
Prompt	one fixed instruction, chat template
Sampling	$k = 64$, $T = 0.8$, top-p 1.0, `max_tokens` 3072
Selection	$s(a) = \log n_a + 0.5\,\bar{z}(a)$, where $\bar{z}$ is the mean within-problem standardised verifier score over the samples that produced a
Verifier	LoRA adapter on the same base: rank 16, α 32, lr 10^{-4}, 1 epoch, 26,034 training rows, neutral judging stance
Ties	broken to the smallest answer

A.3 Verifier training data

The reward model is trained on the base model's own generations over the organiser's training split, labelled by the gold answer. Three choices matter more than the hyperparameters.

Only problems with both a correct and an incorrect sample are used. Training on the unfiltered set teaches the model that hard-looking problems are wrong, which raises overall AUC and does nothing for the within-problem ranking that decides an answer. Each problem contributes an equal number of correct and incorrect solutions, at most three per side. Problems flagged as label-suspect are excluded, because an incorrect gold label makes a correct solution a negative example.

A.4 Compute

Stage	unit	time
Generation, $k = 64$	1,000 problems	44 min
Verifier scoring	1,000 problems $\times$ 64	12 min
Adapter training	one run	50 min to 1 h 50 min
Full submission pipeline	831 problems	56 min
Project total		$\approx$ 40 GPU-hours

External API calls, used only to build candidate training data on training-split problems, cost about \$22 and no local compute. No test problem was sent to any external service at any point.

A.5 What is released

Training and inference code; ten LoRA adapters, one per training run including the seven that failed; the verifier training data; every generation file behind the numbers in this paper; the validation split definition; the preregistration record; and the environment snapshot above. The merged full-precision models are not released — each is reconstructible from its adapter and the base in a few minutes.

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